Project Report: AI-Powered Personal Spending Tracker

Course: Fundamentals of AIML

Student Name: Aditya Namdev

Registration no.:25BCE10297

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1. Introduction

In today's fast-paced student life, managing daily expenses has become an important skill. Many students spend money on food, travel, and personal needs without properly tracking their expenses. This often leads to overspending and poor financial habits.

This project focuses on developing a simple **AI-based expense tracking system** that helps users monitor their daily spending and provides intelligent suggestions to improve financial discipline.

The system is based on **rule-based artificial intelligence**, where decisions are made using predefined logic instead of machine learning.

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2. Problem Statement

Students often face difficulty in managing their daily expenses due to lack of awareness and tracking. They do not realize where most of their money is being spent.

The problem is:

- No proper tracking of expenses
- Lack of categorization
- No feedback or suggestions on spending habits

This project aims to solve these issues by creating a system that:

- Tracks expenses
- Categorizes spending automatically
- Provides useful advice

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3. Objective

The main objectives of this project are:

- To develop a system that records daily expenses
- To automatically categorize expenses using keywords
- To analyze spending patterns
- To provide intelligent suggestions using AI logic
- To help users stay within their daily budget

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4. Methodology

The project follows a simple and structured approach:

Step 1: Input Collection

The user enters:

- Expense description (e.g., "zomato order", "bus ticket")
- Expense amount

Step 2: Category Detection

The system uses **pattern matching** to detect the category:

- Food
- Travel
- Academic
- Personal
- Other

Step 3: Processing

The system calculates:

- Total spending
- Comparison with daily budget

Step 4: Decision Making

Using **IF-ELSE rules**, the system generates advice.

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5. AI Concepts Used

This project uses basic Artificial Intelligence concepts:

◈ Rule-Based System

The system follows predefined rules to make decisions.

◈ Pattern Matching

Keywords in the description are matched to identify categories.

◈ Decision Making

Conditions are checked using IF-ELSE logic to generate advice.

Example:

- If total spending exceeds budget → show warning
- If food spending is high → suggest reducing it

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6. Implementation

The project is implemented using **Python**.

Key Components:

- **Category Dictionary** → stores keywords for classification
- **Function: spending_category()** → identifies category
- **Function: expenditure()** → applies AI rules
- **Main Loop** → interacts with user

Sample Rule:

- If total spending > daily budget → show critical alert
- If food expense > ₹250 → suggest saving

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7. Features

- Automatic expense categorization
- Real-time spending tracking
- Budget monitoring

- AI-based suggestions
- Simple command-line interface

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8. Challenges Faced

- Choosing appropriate keywords for categories
- Designing meaningful AI rules
- Handling user input correctly
- Keeping the system simple but effective

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9. Limitations

- No data storage (data resets after program ends)
- Limited keyword-based categorization
- No graphical interface
- Fixed daily budget

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10. Future Improvements

- Add file/database storage
- Develop GUI using Tkinter
- Add charts and graphs
- Improve category detection using advanced techniques

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11. Conclusion

This project successfully demonstrates how basic AI concepts can be used to solve real-life problems. The system helps users understand their spending habits and encourages better financial decisions.

Even without machine learning, the project shows that **rule-based intelligence can still be powerful and useful** when applied correctly.